Course: BM20A4703 Partial Differential Equations with Applications.
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Worksheet 1

Exercise 1.1

Let

$$\Phi(x,t) = \begin{cases} \frac{1}{(4\pi t)^{n/2}} e^{-\frac{|x|^2}{4t}}, & x \in \mathbb{R}^n, t > 0 \\ 0, & x \in \mathbb{R}^n, t = 0. \end{cases}$$

(1) [5] Prove that $\Phi \in C^\infty(\mathbb{R}^n \times (0,\infty))$.
(2) [5] Prove that, for each $t > 0$,

$$\int_{\mathbb{R}^n} \Phi(x,t)\mathrm{d}x = 1.$$

Hint: You may use the Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2}\mathrm{d}x = \sqrt{\pi}$.
(3) [5] Verify that Φ solves the heat equation

$$u_t - \Delta u = 0 \text{ in } \mathbb{R}^n \times (0,\infty).$$

(4) [5] Given $g \in C(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$. Let u be defined as the *convolution* of $\Phi(\cdot,t)$ and g

$$u(x,t) = (\Phi(\cdot,t) * g)(x) := \int_{\mathbb{R}^n} \Phi(x-y,t)g(y)\mathrm{d}y \tag{1}$$

Prove that $u \in C^\infty(\mathbb{R}^n \times (0,\infty))$ and that u solves the heat equation in $\mathbb{R}^n \times (0,\infty)$. (This verifies a comment made in the lecture: the solution to the heat equation instantly becomes smooth!)
(5) [10] Prove that u given in (1) indeed "attains" its initial value at g, that is, for each $x_0 \in \mathbb{R}^n$,

$$\lim_{x \to x_0, t \to 0^+} u(x,t) = g(x_0).$$

Hint: Use (2) and the definition of limit.

Exercise 1.2

[10] Let u be given by (1). Prove that if g is continuous, non-negative and not identically zero ($g \not\equiv 0$) then $u(x,t) > 0$ for all $x \in \mathbb{R}^n$ and $t > 0$. Carefully explain your answer!

Exercise 1.3

Let u be a smooth solution of the heat equation

$$u_t - \Delta u = 0 \text{ in } \mathbb{R}^n \times (0,\infty)$$

For each $\lambda \in \mathbb{R}$, denote $u_\lambda(x,t) := u(\lambda x, \lambda^2 t)$.
(1) [5] Prove that u_λ solves the heat equation.
(2) [5] Let $w(x,t) := x \cdot Du(x,t) + 2tu_t(x,t)$. Show that w also solves the heat equation. Hint: Use (1).

Exercise 1.4

[10] Verify in detail that, for $x \in \mathbb{R}^n, t > 0$,

$$\mathcal{F}^{-1}\left(e^{-t|\xi|^2}\right) := \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} e^{ix\cdot\xi} e^{-t|\xi|^2}\mathrm{d}\xi = \frac{1}{(2t)^{n/2}} e^{\frac{-|x|^2}{4t}}.$$

Hint: You may use Fubini's theorem.